

Week 5 (CS 418 @ UIC)

Week 5 Slide Deck

Visualizations

Jack Bandy 2026

Topic Title Placeholder

CS418 Week 5 ● Clark/Lake ●





Demo Content Slide

Placeholder content for Week 5.

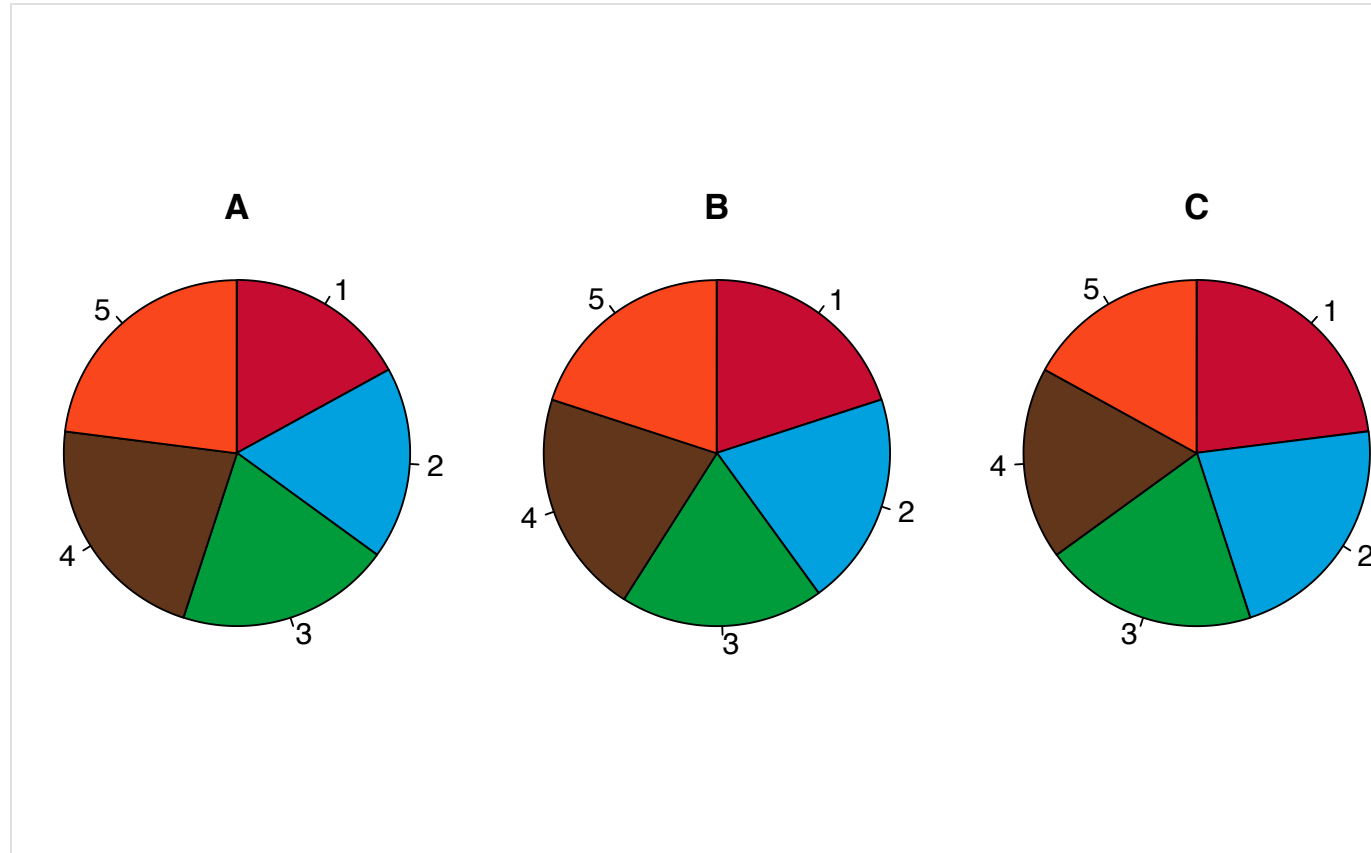
- Visualization goals
- Encodings
- Chart selection

Pie & Donut Charts

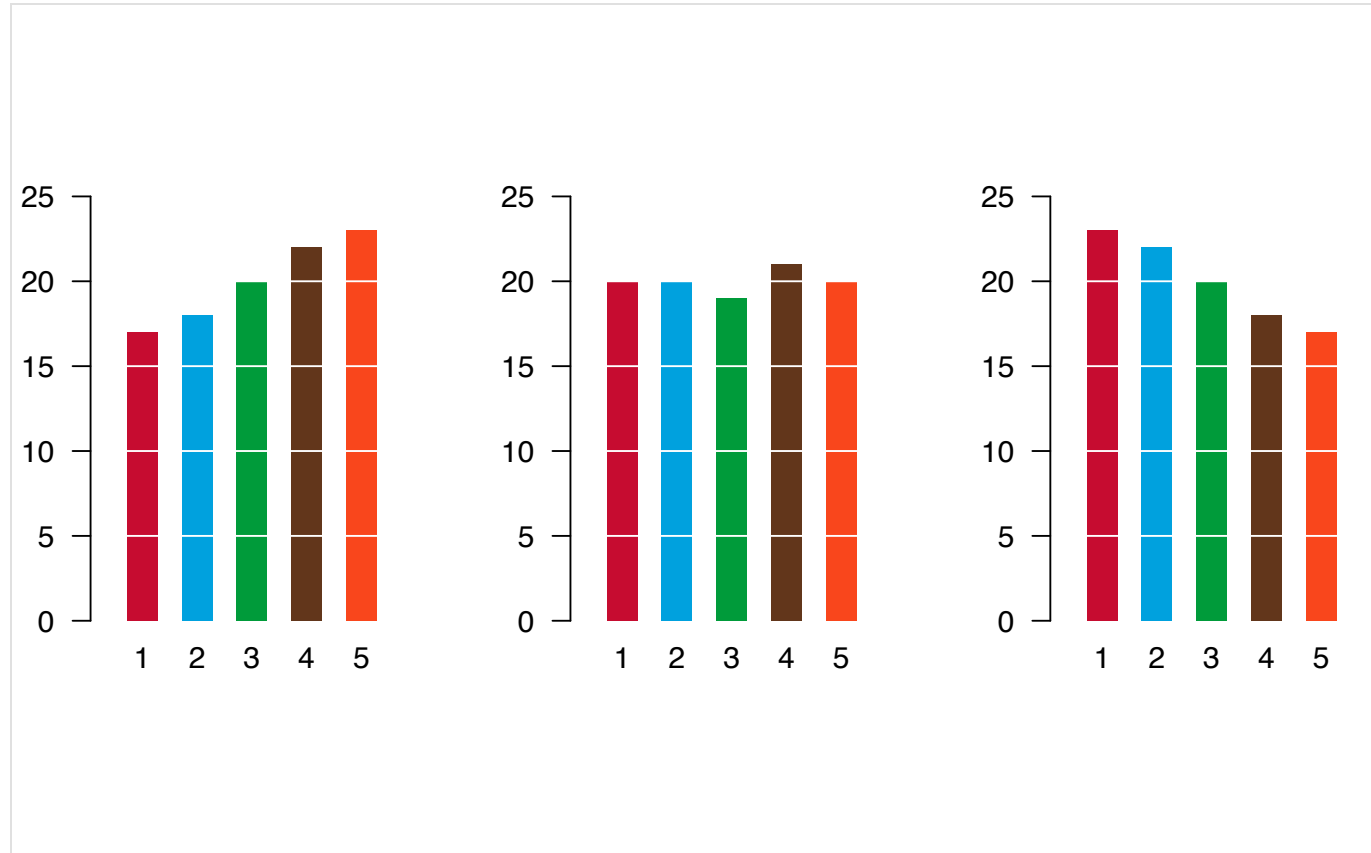
Pie Charts Intuition

- A pie chart encodes **proportion** as **angle** (and area)
- Works okay for “roughly half” or “roughly a quarter”
- Difficult to compare multiple slices

Pie vs. Bar Example



Pie vs. Bar Example



The Donut Chart

- A **donut chart** is a pie chart with the center cut out
- Maintain the “this is all part of a whole” message
- Area comparison

Donut in the Wild: FIFA World Cup 2026



Recreating the FIFA Chart: Setup

```
1 import matplotlib.pyplot as plt
2
3 sizes = [63, 27, 10]
4 labels = ['COL', 'COD', 'In Contest']
5 colors = ['#f5c800', '#0057a8', '#00a896']
6
7 fig, ax = plt.subplots(figsize=(4, 4))
```

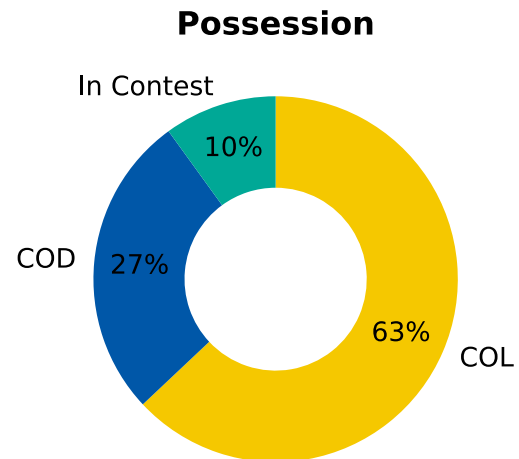
Recreating the FIFA Chart: Draw

```
1 ax.pie(sizes, labels=labels, autopct='%1.0f%%',  
2       colors=colors, startangle=90, counterclock=False,  
3       wedgeprops=dict(width=0.5), pctdistance=0.75)  
4 ax.set_title('Possession', fontweight='bold')  
5 plt.tight_layout()
```

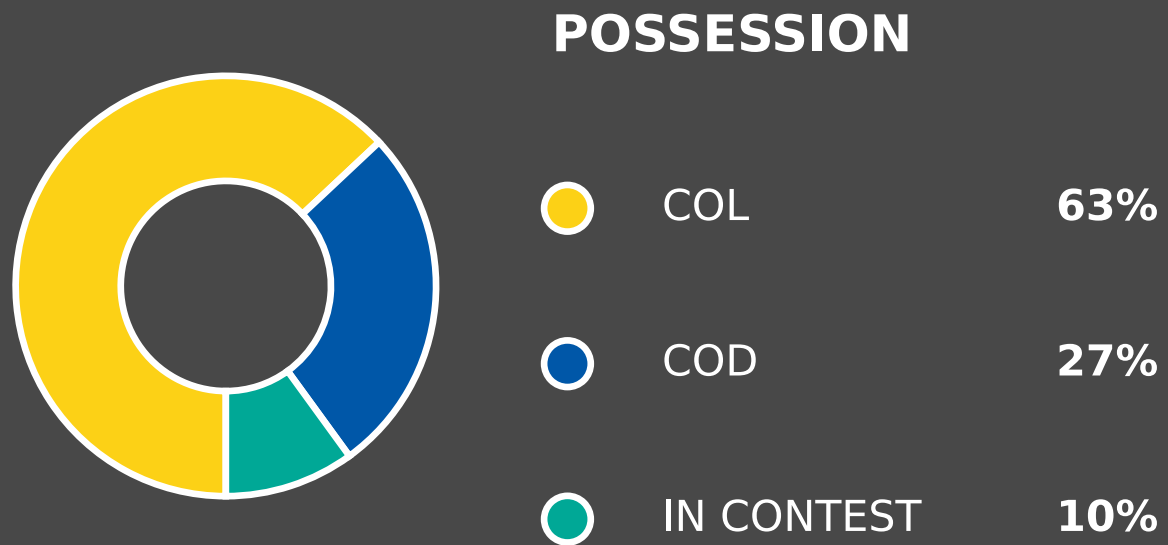
<Figure size 480x296.64 with 0 Axes>

Recreating the FIFA Chart

```
1 import matplotlib.pyplot as plt
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3 sizes = [63, 27, 10]
4 labels = ['COL', 'COD', 'In Contest']
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6 _, ax = plt.subplots()
7 ax.pie(sizes, labels=labels, autopct='%1.0f%%', colors=colors,
8       startangle=90, counterclock=False,
9       wedgeprops=dict(width=0.5), pctdistance=0.75)
10 ax.set_title('Possession', fontweight='bold');
```



Recreating the FIFA Chart: Polished



Sources

1. GitHub source: <https://github.com/jackbandy/data-science-fun/blob/main/docs/slides/week5.qmd>.
2. Last modified and compiled June 24, 15:02h Central (Chicago) time.
3. Pie vs. bar comparison: Schutz, *Piecharts.svg*, Wikimedia Commons, 2008. [CC BY 1.0](#).
4. Slides developed using materials from [Elena Zheleva](#) and [Gonzalo Bello Lander](#), the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart.
5. Slide deck built with [Quarto](#) revealjs.
6. Title font is Big Shoulders; Body font is [Libre Franklin](#).